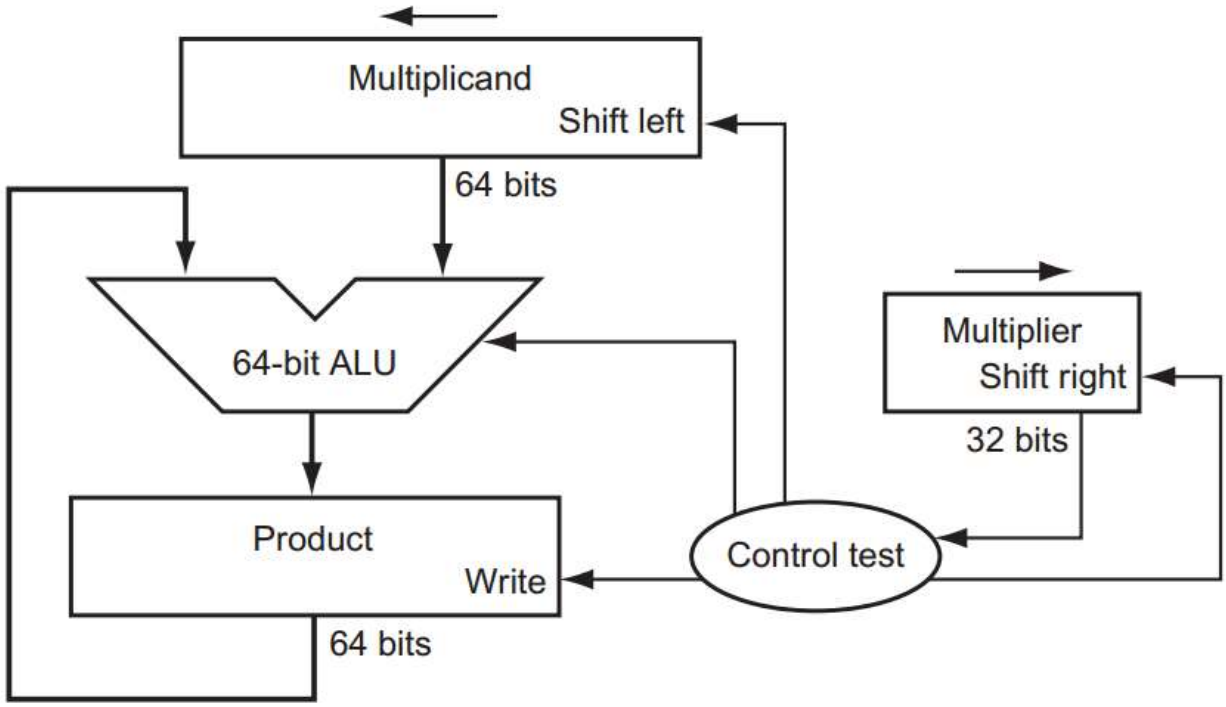
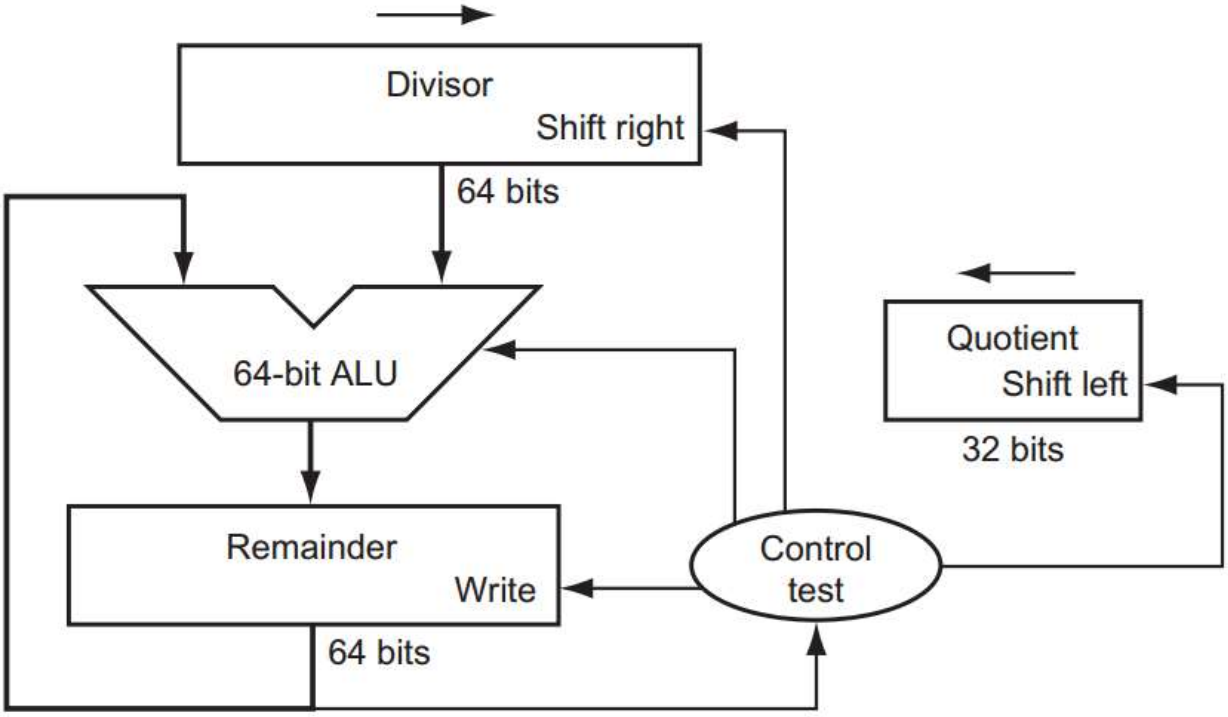
Homework 3

1. What is 5ED4 – 07A4 when these values represent unsigned 16-bit hexadecimal numbers? The result should be written in hexadecimal. Show your work.
What is 5ED4 – 07A4 when these values represent signed 16-bit hexadecimal numbers stored in sign-magnitude format? The result should be written in hexadecimal. Show your work.



2. Using a table similar to that shown in the Figure 3.3 (in text book), calculate the product of the hexadecimal unsigned 6-bit integers 62 and 12 using the hardware described in the above Figure. You should show the contents of each register on each step



Using a table similar to that shown in Figure 3.10 (in text book), calculate 74 divided by 21 using the hardware described in above Figure. You should show the contents of each register on each step. Assume both inputs are unsigned 6-bit integers.

3. What decimal number does the bit pattern 0×0C000000 represent if it is a two's complement integer? An unsigned integer?
If the bit pattern 0×0C000000 is placed into the Instruction Register, what MIPS instruction will be executed?
What decimal number does the bit pattern 0×0C000000/0x172E783A/0x0A34848F represent if it is a floating-point number? Use the IEEE 754 standard

4. Write down the binary representation of the decimal number 63.25, 30.5438×10^{5}, 43.374×10^{-4} assuming the IEEE 754 single precision format.
Write down the binary representation of the decimal number 63.25, 30.5438×10^{5}, 43.374×10^{-4} assuming the IEEE 754 double precision format

5. Calculate the sum of 2.6125×10^{1} and $4.150390625 \times 10^{-1}$ by hand
Calculate the product of -8.0546875×10^{0} and $-1.79931640625 \times 10^{-1}$ by hand